Faculty Vitae: Anh Ngo

1. Education

B.S. Theoretical Physics, Hanoi University of Science, Vietnam National University, 2000
M.S. The ICTP Advanced Diploma Theoretical Condensed Matter Physics, ICTP, Trieste, *Italy*. 2002
PhD Department of Physics and Astronomy, Ohio University, USA, 2010

2. Academic Experience

Professor : Department of Chemical Engineering, University of Illinois- Chicago and Joint Appointment with Argonne National Laboratory : August 2025 – present.

Associate Professor : Department of Chemical Engineering, University of Illinois- Chicago and Joint Appointment with Argonne National Laboratory : August 2020 – 2025.

Staff scientist: Materials Science Division, Argonne National Laboratory, March 2017 — 2020.

Postdoctoral Fellow: Materials Science Division and Advanced Photon Source, Center for Nanoscale Materials, Argonne National Laboratory, March 2014 — February 2017.
Postdoctoral Fellow: Department of material sciences, University of Wisconsin-Madison, USA, May 2012– March 2014.
Postdoctoral Fellow: Department of Physics and Astronomy, Louisiana State University, USA, 2010— May 2012

3. Non-Academic Experience

Staff scientist: Materials Science Division, Argonne National Laboratory, March 2017 — 2020.

Postdoctoral Fellow: Materials Science Division and Advanced Photon Source, Center for Nanoscale Materials, Argonne National Laboratory, March 2014 — February 2017.

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers

6. Honors and awards

Not applicable

7. Service activities

Not applicable

8. Selected Publications

1. Selva Chandrasekaran Selvaraj, Volodymyr Koverga, **Anh T. Ngo**. Exploring Li-Ion transport properties of Li3TiCl6: a machine learning molecular dynamics study. J. Electrochem. Soc. 171 050544. (2024).
2. Guo-Xing Li, et, al. , **Anh T. Ngo**, Donghai Wang. Enhancing lithium-metal battery longevity through minimized coordinating diluent. **Nature Energy** 9, 817–827 (2024).
3. Weiran Zhang, et, al. **Anh T. Ngo**, Chunsheng Wang. Single-phase local-high-concentration solid polymer electrolytes for lithium-metal batteries. **Nature Energy** 9, 386–400 (2024).
4. Guo-Xing Li, et, al. , **Anh T Ngo**, Gunther Brunklaus, Donghai Wang. Interfacial solvation-structure regulation for stable Li metal anode by a desolvation coating technique. **Proceedings of the National Academy of Sciences** 121 (4), e2311732121, (2024).
5. Tolulope M Ajayi, et, al. , **Anh T. Ngo**, Saw-Wai Hla. Characterization of just one atom using synchrotron X-rays. **Nature** 618, 69–73 (2023).
6. Jijian Xu, Volodymyr Koverga, et, al., **Anh T. Ngo**, Chunsheng Wang. Revealing the anion–solvent interaction for ultralow temperature lithium metal batteries. Advanced Materials 36 (7), 2306462, (2024).
7. Xiaozhao Liu, Volodymyr Koverga, Hoai T. Nguyen, **Anh T. Ngo**, Tao Li. Exploring solvation structure and transport behavior for rational design of advanced electrolytes for next generation of lithium batteries. Appl. Phys. Rev. 11, 021307 (2024).
8. Huu Do, Alex Lee Hyowon Park and **Anh T. Ngo**, Delocalized polaron and Burstein-Moss shift induced by Li in α−V2O5 : A DFT+DMFT study. Physical Review B 108 (20), 205122, (2023).
9. Alex Lee, Hyowon Park and **Anh T. Ngo.** Effect of off-diagonal elements in the Wannier Hamiltonian on DFT+DMFT for low-symmetry materials: Study of Li2MnO3. Phys. Rev. B 108, 205146, (2023).
10. Yue Gao, Tomas Rojas, Ke Wang, Shuai Liu, Daiwei Wang, Tianhang Chen, Haiying Wang, **Anh T. Ngo**, Donghai Wang. " Low-temperature and high-rate-charging lithium metal batteries enabled by an electrochemically active monolayer-regulated interface". **Nature Energy** 5, 534–542 (2020).
11. Pallab Barai, **Anh Ngo**, Badri Narayanan, Kenneth Higa, Larry A Curtiss, Venkat Srinivasan." The role of local inhomogeneities on dendrite growth in LLZO-based solid electrolytes" Journal of The Electrochemical Society, 167 100537 (2020).
12. Hyowon Park, Ravindra Nanguneri, and **Anh T. Ngo.**"DFT DMFT study of spin-charge-lattice coupling in covalent LaCoO3" Phys. Rev. B 101, 195125 (2020).
13. Zhe Zhang, Yiming Li, Bo Song, Yuan Zhang, Xin Jiang, Ming Wang, Ryan Tumbleson, Changlin Liu, Pingshan Wang, Xin-Qi Hao, Tomas Rojas, **Anh T. Ngo**, Jonathan L Sessler, George R Newkome, Saw Wai Hla, Xiaopeng Li . " Intra-and intermolecular self-assembly of a 20-nm-wide supramolecular hexagonal grid" **Nature Chemistry** 12, 468–474 (2020).
14. Yuan Zhang, Jan Patrick Calupitan, Tomas Rojas, Ryan Tumbleson, Guillaume Erbland, Claire Kammerer, Tolulope Michael Ajayi, Shaoze Wang, Larry A Curtiss, **Anh T. Ngo**,

Sergio E Ulloa, Gwénaël Rapenne, Saw Wai Hla. “ A chiral molecular propeller designed for unidirectional rotations on a surface” **Nature Communications** 10, 3742 (2019).

15. Yipeng Cao, Milind J Gadre, **Anh T. Ngo**, Stuart B Adler, Dane D Morgan. “Factors controlling surface oxygen exchange in oxides” **Nature Communications** 10, 1346 (2019).
16. M. Asadi, B. Sayahpour, P. Abbasi, **Anh T. Ngo**, Klas Karis, Jacob R Jokisaari, C. Liu, B. Narayanan, Marc Gerard, Poya Yasaei, X. Hu, A. Mukherjee, Kah Chun Lau, R. S Assary, Fatemeh Khalili-Araghi, Robert F Klie, Larry A Curtiss and Amin Salehi-Khojin. “A lithium–oxygen battery with a long cycle life in an air-like atmosphere”. **Nature** 555, 502–506 (2018).
17. Yang Li, **Anh T. Ngo**, Andrew DiLullo, Kyaw Zin Latt, Heath Kersell, Brandon Fisher, Peter Zapol, Sergio E Ulloa, Saw-Wai Hla. “ Anomalous Kondo resonance mediated by semiconducting graphene nanoribbons in a molecular heterostructure ” **Nature Communications** 8, 946 (2017).
18. **Anh T. Ngo**, Eugene H. Kim, and Sergio E. Ulloa “Single-atom gating and magnetic interactions in quantum corrals” Phys. Rev. B 95, 161407(R) (2017).
19. Philippe Debray, SMS Rahman, J Wan, RS Newrock, M Cahay, **Anh T. Ngo**, SE Ulloa, ST Herbert, Mustafa Muhammad, M Johnson “ All-electric quantum point contact spin-polarizer” **Nature Nanotechnology** 4, 759–764 (2009).

9. Professional Development

Not applicable